

## Chapter 134 — Continent Urinary Diversion — Numbers, Gold Standards and Traps

### IF YOU READ NOTHING ELSE

- **Commonest diversion worldwide** → ileal conduit (Bricker, 1950).
- **Neobladder candidates**: 80–90% men, ~75% women.
- **Renal cut-off**: estimated glomerular filtration rate (eGFR) >35–40 mL/min or creatinine <1.7–2.2 mg/dL.
- **Urethral frozen section** = definitive test, both sexes (American Urological Association, AUA, requires negative margin).
- **Male urethral recurrence risk** → prostatic stromal invasion.
- **Female urethral risk** → bladder-neck / anterior vaginal tumour; no skip lesions.
- **Mature pouch** ≥500 mL; 44 cm ileum <200 mL → 500–600 mL by 1 year.
- **Antireflux unnecessary**: Studer vs T pouch randomised controlled trial (RCT) — T pouch more reoperations.
- **Night incontinence 20–50%**; day continence 80–90%.
- **Retention**: men 4–10%, women 20–30%; treat with self-catheterisation.

### HISTORY

- **First bowel diversion**: Simon 1852, ureterosigmoidostomy, exstrophy.
- **Ureterosigmoidostomy late risks**: hyperchloraemic acidosis, pyelonephritis, cancer at implant site.
- **Ileal conduit**: Zaayer 1911 → Bricker 1950.
- **Gilchrist 1950**: ileocaecal valve continence, first human continent cutaneous.
- **Kock 1982**: intussuscepted nipple; established detubularisation + double folding.
- **Tizzoni-Poggi 1888**: first orthotopic (dog). **Lemoine 1913**: first human.
- **Camey-Le Duc 1979**: tubular ileum → high pressure, poor continence.
- **Women**: continence via urethra alone, mid-1990s — bladder neck not needed.
- **RAZOR** (extracorporeal): 21% continent. **iROC** (intracorporeal): 11%.
- **Guideline (AUA/Society of Urologic Oncology, European Association of Urology)**: counsel on all three diversion options.

### BASIC PRINCIPLES

- **Three principles**: reliable outlet; low pressure (detubularise, sphere); ≥500 mL.
- **Sphere** → maximum volume per surface area.
- **Ileum stretches more than colon**.

### PATIENT SELECTION

- **Absolute contraindication (CI), orthotopic**: invasive cancer at urethral margin; significant urethral stricture.
- **Absolute CI, all continent**: severe renal/hepatic dysfunction; cannot self-catheterise.
- **Unplanned diversion change**: up to ~10% → always mark conduit site.
- **Stoma**: through middle of rectus → avoids parastomal hernia.
- **Algorithm (Fig 134.2)**: eGFR >40 → sphincter → bowel → not frail → tumour check.
- **Male urethral recurrence**: ~7% 5 yr, ~9% 10 yr; median 2 yr.
- **Risk order**: no prostate < carcinoma in situ (CIS)/ductal < stromal invasion.
- **Bladder CIS**: not a predictor (either sex).
- **Neobladder protective** vs urethral recurrence: hazard ratio (HR) 0.44 (meta-analysis).
- **Transurethral resection (TUR) prostatic biopsy**: 5 and 7 o'clock beside verumontanum; not mandatory.
- **Stromal invasion** → neoadjuvant chemotherapy (usually node-positive).
- **Persistent prostatic urethral disease** → cutaneous diversion + urethrectomy.

- **Women**: negative bladder neck → negative urethra (no skip lesions).
- **Female assessment**: bimanual examination under anaesthesia; palpable neck/vaginal tumour → urethrectomy + cutaneous.
- **Local pelvic recurrence** <10% (~7%); rarely affects neobladder.
- **Do not operate orthotopic if**: gross positive margin at pubis, rectum or side wall.
- **pT3/N+ at surgery** → not a contraindication.
- **Age, obesity** → not contraindications.
- **Commonest technical conversion cause**: short/fatty mesentery.
- **Self-catheterisation**: 10–20% men, ~30% women, 100% cutaneous.
- **Poor candidates**: multiple sclerosis, quadriplegia, frail, cognitively impaired → conduit.
- **Weak male sphincter** → artificial urinary sphincter; **female stress incontinence** → Burch/sling (then intermittent self-catheterisation, ISC).
- **Post-radiation neobladder** if no incontinence, stricture, fistula or severe bowel damage.
- **Worst sphincter scarring**: prostate brachytherapy/high-dose radiotherapy, cervical vaginal implant.
- **Post-prostatectomy**: continent before → continent after.
- **Prior bowel resection**: reuse old anastomosis; extra 45–60 cm → short bowel risk.

### CONTINENCE PRESERVATION

- **Rhabdosphincter**: omega-shaped, anterolateral; pudendal nerve.
- **Female continence zone**: mid-urethra.
- **Male**: urethral sutures before division (urethra retracts); absorbable dorsal venous complex (DVC) suture.
- **Female**: urethra above endopelvic fascia, no retraction → suture after removal.
- **Apical tumour** → urethrectomy + cutaneous/conduit; focal CIS → more distal margin.
- **Female**: omental flap between vagina and neobladder; sacrocolpopexy.
- **Vaginal dissection**: not dorsal to 2 and 10 o'clock.

### TECHNIQUES FOR CONTINENT BLADDER SUBSTITUTION

- **Gold standard segment**: ileum (best compliance, mobile mesentery, mucosal atrophy).
- **Worst compliance**: stomach and sigmoid.
- **Stomach**: secretes acid → option in renal impairment; dysuria, haematuria.
- **Distal ileum loss** → vitamin B12 deficiency (Indiana).
- **Le Duc stricture** 7–29% vs direct anastomosis <3%.

- **Ureteroileal stricture lowest:** Leadbetter/Wallace, interrupted absorbable, 3–6%.
- **Hautmann reflux switch (1997):** stenosis 9.5% → 1%.
- **Studer reflux** visible at ~300 mL; kidneys preserved ~95%.
- **Kock nipple late failure:** stenosis, extussusception, staple stones, >10 yr, silent.
- **Suture:** absorbable only; metal staples → stones.
- **Ileal reservoirs:** 60–75 cm terminal ileum.
- **Studer:** 40–44 cm reservoir + afferent limb; isolate 15–20 cm from ileocaecal valve.
- **Hautmann:** ~70 cm W; larger capacity → earlier continence, more retention.
- **Serous-lined tunnel:** 40 cm W; needs long, non-dilated ureters.
- **T pouch:** 15-cm afferent limb; accepts dilated or short ureters.
- **Colon neobladder:** diseased/resected ileum; needs larger initial volume.
- **Enhanced Recovery After Surgery (ERAS):** no bowel prep (except colon), no nasogastric tube, alvimopan.
- **Commonest gastrointestinal complication:** ileus; alvimopan reduces it.
- **Malnourished/frail:** up to 70% of cystectomy patients.
- **Catheter out** at 3 weeks; no routine pouchogram.
- **Umbilical stoma:** least stenosis, easiest for paraplegics; medical alert bracelet.
- **Appendix channel:** ≥5–6 cm, ≥12 Fr; takes 14–16 Fr only.
- **Mitrofanoff 1980:** tunnelled appendix; **Mitchell:** caecal lengthening.
- **Yang-Monti:** 2–3 cm ileum retubularised transversely.
- **Indiana:** double Lemberg imbrication of valve, 3–4 cm, over 16 Fr.
- **Right colon pouch:** ~30 cm colon; ureteroileal; ileocaecal valve = antireflux.
- **Test channel:** fill, compress, recatheterise before closure.
- **Undrained leak** → percutaneous drain / nephrostomies, not early open repair.
- **Efferent necrosis below fascia** → explore.
- **Stomal stenosis** up to 20%; late needs open revision.
- **Treat bacteriuria only** if symptomatic or Proteus/Pseudomonas.
- **Late febrile urinary infection** → exclude obstruction, stones, poor emptying.
- **Pouch perforation:** cystogram; stable → catheter drainage.
- **Endoscopic stricture treatment:** retreatment >50%; open repair better.
- **Pouch–vaginal fistula:** 5–10%; lateral cystogram before catheter out.
- **Daytime incontinence risks:** age >65, colon, no nerve sparing, diabetes.
- **Continence plateau:** 12–18 months before intervening.
- **USC prospective:** day 92% (12–18 mo), night 51% (18–36 mo).
- **Female questionnaire:** ~60% self-catheterise daily.
- **Retention risk (men):** excess bowel length, prostate sparing.
- **Male retention-free:** 93% 3 yr, 88% 5 yr, 79% 10 yr.
- **Female retention:** mechanical (posterior angulation), not neurogenic.
- **Sling/injection:** never blind needles; bone anchors or prepubic route.

## FOLLOW-UP

- **Phases:** <4 mo strictures; 4 mo–3 yr cancer; >3 yr stones, obstruction, urethral cancer.
- **Visits:** 4-monthly yr 1 → 6-monthly to 3 yr → annual.
- **Annual:** B12; urethral wash if CIS; prostate-specific antigen if prostate cancer.
- **After 5 yr:** annual renal ultrasound, indefinitely.
- **Women:** bimanual + voided cytology (urethral recurrence invisible on CT).
- **Health-related quality of life:** no diversion proven superior; neobladder → better body image.

## RESULTS AND COMPLICATIONS

- **Early/late cut-off:** 90 days.
- **Early morbidity:** neobladder = conduit; cutaneous → more infection, dehiscence.

## CLASSIC TRAPS

- **Trap:** old age excludes neobladder — **Actually:** age alone never a contraindication.
- **Trap:** bladder CIS bars orthotopic — **Actually:** negative urethral frozen section suffices.
- **Trap:** preoperative prostatic TUR biopsy mandatory — **Actually:** frozen section decides.
- **Trap:** women need bladder neck for continence — **Actually:** rhabdosphincter, mid-urethra.
- **Trap:** antireflux protects kidneys — **Actually:** obstruction risk; refluxing Studer equal.
- **Trap:** obesity forbids neobladder — **Actually:** may suit better than a stoma.
- **Trap:** treat all neobladder bacteriuria — **Actually:** only symptomatic or Proteus/Pseudomonas.
- **Trap:** early open repair of urine leak — **Actually:** percutaneous drainage first.
- **Trap:** women mainly leak — **Actually:** women mainly retain; men leak.
- **Trap:** "IC" in Tables 134.1/134.2 = ileal conduit — **Actually:** intermittent catheterisation.